
This document provides additional assistance with wiring your Extron IP Link Pro Control Processor to your device. Different components may require a different wiring scheme than those listed below.

For complete operating instructions, refer to the user's manual for the specific IP Link Pro Control Processor or the documentation supplied by the manufacturer of the controlled device.

For more information on using Global Scriptor Modules, refer to the "[Guide to Using Scriptor Modules](#)" document.

Device Specifications

Device Type: Video Projector
Manufacturer: Christie
Firmware Version: N/A
Model(s): Roadster S+16K, Roadster S+12K, Roadster S+20K

Tested on the Following Software and Firmware Versions

IP Link Pro Control Processor Firmware	Global Scriptor Version
3.00.0000-b022	2.0.0

Version History

Module Version	Date	Notes
1_0_0_0	10/15/2018	Initial Version

Module Notes

- Unidirectional variable must be set to 'True' if status is not required. Default value is 'False'.
Example: `InterfaceName.Unidirectional = 'True'`
- connectionCounter variable must be set to the number of queries that will be sent to the device before displaying 'Disconnected' if no response is received. Default value is 15.
Example: `InterfaceName.connectionCounter = 5`

Supported Classes and Examples

SerialClass
<code>InterfaceName = ModuleName.SerialClass(ProcessorName, 'COM1', Model='Roadster S+16K')</code>
SerialOverEthernetClass
<code>InterfaceName = ModuleName.SerialOverEthernetClass('192.168.254.254', 2001, Model='Roadster S+16K')</code>
EthernetClass
<code>InterfaceName = ModuleName.EthernetClass('192.168.254.254', 3002, Model='Roadster S+16K')</code>

Control Commands

Format with Qualifier:

```
InterfaceName.Set(Command, Value, {'Qualifier Key': 'Qualifier Value'})
```

Format without Qualifier:

```
InterfaceName.Set(Command, Value)
```

Command AutoImage	Value None		
# AutoImage example InterfaceName.Set('AutoImage', None)			
Command Channel	Value '1' – '50'		
# Channel example InterfaceName.Set('Channel', '1')			
Command Freeze	Value 'On'	Value 'Off'	
# Freeze example InterfaceName.Set('Freeze', 'On')			
Command Input	Value 'BNC' 'S-Video' 'Slot 2 (Input 1)' ¹	Value 'DVI-I' 'Slot 1 (Input 1)' ¹ 'Slot 2 (Input 2)' ¹	Value 'Composite' 'Slot 1 (Input 2)' ¹
# Input example InterfaceName.Set('Input', 'BNC')			
Command Keypad	Value '0' – '9'		
# Keypad example InterfaceName.Set('Keypad', '1')			
Command LampMode	Value 'Maximum Brightness'	Value 'Maintain Intensity'	Value 'Maintain Power'
# LampMode example InterfaceName.Set('LampMode', 'Maximum Brightness')			
Command MenuNavigation	Value 'Up' 'Right' 'Exit'	Value 'Down' 'Enter'	Value 'Left' 'Menu'
# MenuNavigation example InterfaceName.Set('MenuNavigation', 'Up')			
Command OnScreenDisplay	Value 'On'	Value 'Off'	

# OnScreenDisplay example InterfaceName.Set('OnScreenDisplay', 'On')			
Command PictureInPicture	Value 'On'	Value 'Off'	
# PictureInPicture example InterfaceName.Set('PictureInPicture', 'On')			
Command PictureInPicturePosition	Value 'Top Right' 'Bottom Right'	Value 'Top Left'	Value 'Bottom Left'
# PictureInPicturePosition example InterfaceName.Set('PictureInPicturePosition', 'Top Right')			
Command PictureInPictureSwap	Value None		
# PictureInPictureSwap example InterfaceName.Set('PictureInPictureSwap', None)			
Command Power	Value 'On'	Value 'Off'	
# Power example InterfaceName.Set('Power', 'On')			
Command TestPattern	Value 'Off' 'White' 'Checker' 'Aspect Ratio'	Value 'Grid' 'Flat Grey' '13 Point' 'Edge Blend'	Value 'Grey Scale' 'Black' 'Color Bars'
# TestPattern example InterfaceName.Set('TestPattern', 'Off')			
Command VideoMute	Value 'On'	Value 'Off'	
# VideoMute example InterfaceName.Set('VideoMute', 'On')			

¹ This Input state requires an optional input board installed in the respective expansion slot.

Status Available

For all commands, call Update to receive the latest status. ConnectionStatus does not support the Update function and is triggered by the device providing a successful response to other Update function calls.

Format with Qualifier:

```
InterfaceName.Update(Command, {'Qualifier Key': 'Qualifier Value'})
Value = InterfaceName.ReadStatus(Command, {'Qualifier Key': 'Qualifier Value'})
InterfaceName.SubscribeStatus(Command, {'Qualifier Key': 'Qualifier Value'}, FeedbackHandler)
FeedbackHandler will be called only when the specified qualifier gets a new status.
```

Format without Qualifier:

```
InterfaceName.Update(Command)
Value = InterfaceName.ReadStatus(Command)
InterfaceName.SubscribeStatus(Command, None, FeedbackHandler)
FeedbackHandler will be called when any qualifier gets a new status.
```

Command	Value		
Channel	'1' – '50'		
# Channel examples InterfaceName.Update('Channel') Value = InterfaceName.ReadStatus('Channel') InterfaceName.SubscribeStatus('Channel', None, FeedbackHandler)			
Command	Value		Value
ConnectionStatus	'Connected'	'Disconnected'	
# ConnectionStatus examples Value = InterfaceName.ReadStatus('ConnectionStatus') InterfaceName.SubscribeStatus('ConnectionStatus', None, FeedbackHandler)			
Command	Value		Value
Freeze	'On'		'Off'
# Freeze examples InterfaceName.Update('Freeze') Value = InterfaceName.ReadStatus('Freeze') InterfaceName.SubscribeStatus('Freeze', None, FeedbackHandler)			
Command	Value		Value
Input	Value		Value
	'BNC'	'DVI-I'	'Composite'
	'S-Video'	'Slot 1 (Input 1)' ¹	'Slot 1 (Input 2)' ¹
	'Slot 2 (Input 1)' ¹	'Slot 2 (Input 2)' ¹	
# Input examples InterfaceName.Update('Input') Value = InterfaceName.ReadStatus('Input') InterfaceName.SubscribeStatus('Input', None, FeedbackHandler)			
Command	Value		Value
LampMode	Value		Value
	'Maximum Brightness'	'Maintain Intensity'	'Maintain Power'
# LampMode examples InterfaceName.Update('LampMode')			

Global Scripter Module Communication Sheet

Value = InterfaceName.ReadStatus('LampMode') InterfaceName.SubscribeStatus('LampMode', None, FeedbackHandler)			
Command LampUsage	Value Hours		
# LampUsage examples InterfaceName.Update('LampUsage') Value = InterfaceName.ReadStatus('LampUsage') InterfaceName.SubscribeStatus('LampUsage', None, FeedbackHandler)			
Command OnScreenDisplay	Value 'On'	Value 'Off'	
# OnScreenDisplay examples InterfaceName.Update('OnScreenDisplay') Value = InterfaceName.ReadStatus('OnScreenDisplay') InterfaceName.SubscribeStatus('OnScreenDisplay', None, FeedbackHandler)			
Command OperationHours	Value Hours		
# OperationHours examples InterfaceName.Update('OperationHours') Value = InterfaceName.ReadStatus('OperationHours') InterfaceName.SubscribeStatus('OperationHours', None, FeedbackHandler)			
Command PictureInPicture	Value 'On'	Value 'Off'	
# PictureInPicture examples InterfaceName.Update('PictureInPicture') Value = InterfaceName.ReadStatus('PictureInPicture') InterfaceName.SubscribeStatus('PictureInPicture', None, FeedbackHandler)			
Command PictureInPicturePosition	Value 'Top Right' 'Bottom Right'	Value 'Top Left'	Value 'Bottom Left'
# PictureInPicturePosition examples InterfaceName.Update('PictureInPicturePosition') Value = InterfaceName.ReadStatus('PictureInPicturePosition') InterfaceName.SubscribeStatus('PictureInPicturePosition', None, FeedbackHandler)			
Command Power	Value 'On' 'No Lamp'	Value 'Off' 'Warming Up'	Value 'Boot Mode'
# Power examples InterfaceName.Update('Power') Value = InterfaceName.ReadStatus('Power') InterfaceName.SubscribeStatus('Power', None, FeedbackHandler)			
Command TestPattern	Value 'Off' 'White' 'Checker'	Value 'Grid' 'Flat Grey' '13 Point'	Value 'Grey Scale' 'Black' 'Color Bars'

	'Aspect Ratio'	'Edge Blend'
<pre># TestPattern examples InterfaceName.Update('TestPattern') Value = InterfaceName.ReadStatus('TestPattern') InterfaceName.SubscribeStatus('TestPattern', None, FeedbackHandler)</pre>		
Command VideoMute	Value 'On'	Value 'Off'
<pre># VideoMute examples InterfaceName.Update('VideoMute') Value = InterfaceName.ReadStatus('VideoMute') InterfaceName.SubscribeStatus('VideoMute', None, FeedbackHandler)</pre>		

¹ This Input state requires an optional input board installed in the respective expansion slot.

Cable and Adapter Requirements

Captive Screw to Female DB9 RS-232 Serial Cable

Notes for the Device

Device also supports RS-422. The pin assignments diagram is shown below

Serial communication

Port Type: RS232, RS422

Baud Rate: 115200

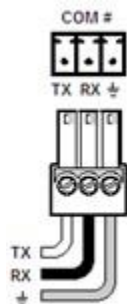
Data Bits: 8

Parity: None

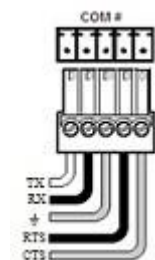
Stop Bits: One

Flow Control: None

Pin Assignments Diagram



Signal	Main Cable	Pin	Signal
TxD	↗	2	TxD
RxD	↖	3	RxD
GND	—	5	GND



Signal	Main Cable	Pin	Signal
RXD-	←	3	TXD-
TXD-	→	2	RXD-
GND	—	7	GND
RXD+	←	5	TXD+
TXD+	→	4	RXD+



Network communication

When configuring the Ethernet module, be sure device settings match those of the Global Scripter ethernet interface

Port Type:	Ethernet
Default Port:	3002
Logon Credentials Supported:	No
Multi-Connection Capabilities:	Undetermined
Port Changeability:	Yes

Ethernet Module Configuration Description

Please refer to user manual for settings and changes to the network communication

Notes for the Device

Appendix A. Set Commands

Auto Image None	(ASU)
Channel 1	(CHA 1)
Channel 50	(CHA 50)
Freeze Off	(FRZ 0)
Freeze On	(FRZ 1)
Input BNC	(SIN 1)
Input Composite	(SIN 3)
Input DVI-I	(SIN 2)
Input Slot 1 (Input 1)	(SIN 5)
Input Slot 1 (Input 2)	(SIN 7)
Input Slot 2 (Input 1)	(SIN 6)
Input Slot 2 (Input 2)	(SIN 8)
Input S-Video	(SIN 4)
Keypad 0	(KEY 0)
Keypad 9	(KEY 9)
Lamp Mode Maintain Intensity	(LPM 1)
Lamp Mode Maintain Power	(LPM 2)
Lamp Mode Maximum Brightness	(LPM 0)
Menu Navigation Down	(KEY 59)
Menu Navigation Enter	(KEY 13)
Menu Navigation Exit	(KEY 27)
Menu Navigation Left	(KEY 60)
Menu Navigation Menu	(KEY 44)
Menu Navigation Right	(KEY 62)
Menu Navigation Up	(KEY 58)
On Screen Display Off	(OSD 0)
On Screen Display On	(OSD 1)
Picture In Picture Off	(PIP 0)
Picture In Picture On	(PIP 1)
Picture In Picture Position Bottom Left	(PPP 2)
Picture In Picture Position Bottom Right	(PPP 3)
Picture In Picture Position Top Left	(PPP 1)
Picture In Picture Position Top Right	(PPP 0)
Picture In Picture Swap None	(PPS)
Power Off	(PWR 0)
Power On	(PWR 1)

Test Pattern 13 Point	(ITP 7)
Test Pattern Aspect Ratio	(ITP 12)
Test Pattern Black	(ITP 5)
Test Pattern Checker	(ITP 6)
Test Pattern Color Bars	(ITP 8)
Test Pattern Edge Blend	(ITP 11)
Test Pattern Flat Grey	(ITP 4)
Test Pattern Grey Scale	(ITP 2)
Test Pattern Grid	(ITP 1)
Test Pattern Off	(ITP 0)
Test Pattern White	(ITP 3)
Video Mute Off	(PMT 0)
Video Mute On	(PMT 1)

Appendix B. Update Commands

Channel	(CHA?)
Freeze	(FRZ?)
Input	(SIN?)
Lamp Mode	(LPM?)
Lamp Usage	(LPH?)
On Screen Display	(OSD?)
Operation Hours	(PJH?)
Picture In Picture	(PIP?)
Picture In Picture Position	(PIP?)
Power	(PWR?)
Test Pattern	(ITP?)
Video Mute	(PMT?)